

BVAR vs. Information Criteria: A Monte Carlo Simulation Study

Akın Aydın* Ferdinand Loser†

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Abstract

This is the abstract for your paper. Briefly state the purpose of the research, the principal results, and major conclusions. The abstract should be self-contained and citation-free.

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*Ludwig-Maximilian University of Munich, akin.aydin@campus.lmu.de

†Ludwig-Maximilian University of Munich, ferdinand.loser@campus.lmu.de

1 Introduction

Introduce the problem of lag selection and parameter proliferation in Structural VARs. Highlight the debate between hard lag selection (Information Criteria) and Bayesian methods (Shrinkage and Model Averaging).

2 Methodology

2.1 Data Generating Process

Explain the true DGP based on Kilian and Murphy (2014) or other literature.

2.2 Optimal Prior Selection via Marginal Data Density

Describe how the shrinkage parameter (τ) is treated as an optimization problem rather than a fixed heuristic.

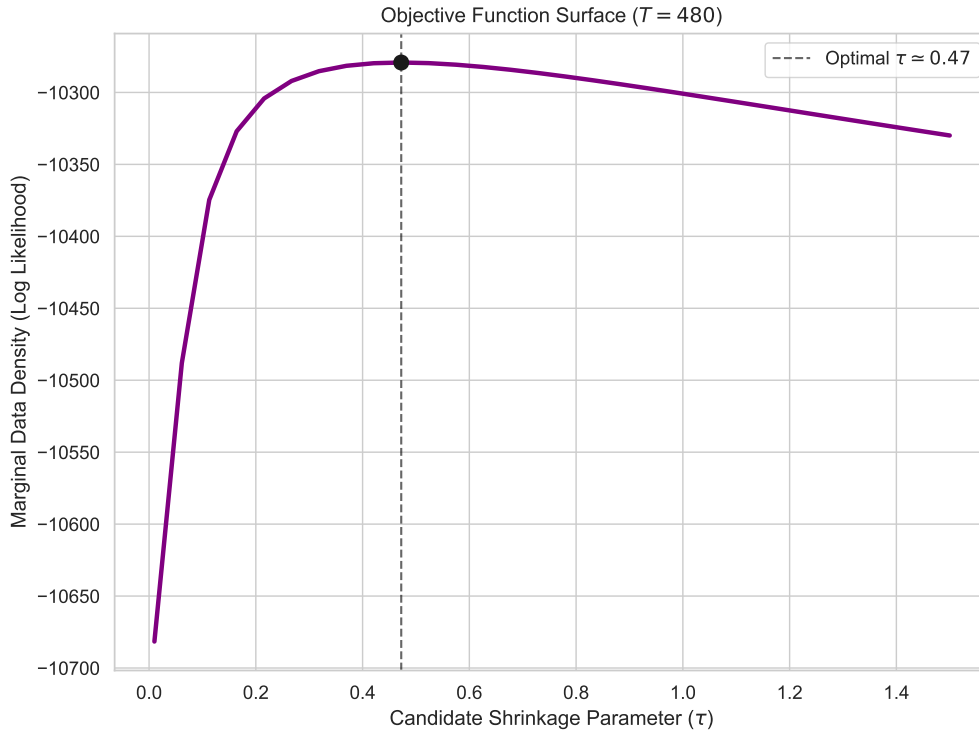


Figure 1: Objective Function Surface for Marginal Data Density Optimization. The peak represents the optimal balance between model fit and parameter penalty.

3 Simulation Results

Present the results of the Monte Carlo simulations, broken down by the specific strategies employed to handle model uncertainty.

3.1 Behavior of Standard Information Criteria

Before evaluating the Bayesian estimators, we examine the baseline lag detection performance of the classical Information Criteria.

Table 1: Average Evaluated Lag Order ($\hat{p}$) selected by standard Information Criteria.

p_0	Estimator T	AIC	AICc	SIC (BIC)	HQC
4	96	9.85	1.00	1.29	1.92
	144	2.65	1.00	1.51	2.02
	240	3.12	1.00	1.98	2.05
	480	3.51	1.00	2.01	2.34
6	96	10.81	1.00	1.22	2.02
	144	3.49	1.00	1.53	2.05
	240	3.97	1.00	1.98	2.04
	480	5.69	1.00	2.00	2.63
8	96	11.21	1.00	1.26	2.26
	144	5.04	1.00	1.51	1.96
	240	6.01	1.00	1.98	2.04
	480	8.01	1.00	2.00	2.68
10	96	11.47	1.00	1.20	2.66
	144	4.86	1.00	1.55	2.03
	240	6.72	1.00	1.96	2.08
	480	8.79	1.00	2.01	2.59

3.2 The Shrinkage Strategy: Priors vs. Data

This section compares hard lag selection against dynamic BVARs. We analyze both the bias-variance tradeoff and how the optimal prior relaxes asymptotically as sample sizes increase.

Table 2: Relative Mean Squared Error: Hard Selection vs. Bayesian Shrinkage.

p_0	Estimator _{Mapped}	T	AIC	AICc	SIC (BIC)	HQC	BVAR-RW ($\tau = 0.05$)	BVAR-RW ($\tau = 0.20$)	BVAR-RW ($\tau = 0.50$)	BVAR-WN ($\tau = 0.20$)	SOTA-BVAR (Opt τ)	SOTA-BMA (Opt τ)
4		96	1.328	0.957	0.985	0.998	190.036	4.063	1.580	1.817	0.929	0.933
		144	1.106	1.206	1.145	1.085	88.872	2.753	1.376	1.655	1.335	1.245
		240	1.052	1.042	1.067	1.056	25.208	1.159	1.017	1.074	0.890	1.020
		480	0.956	1.589	1.214	1.091	9.532	1.150	0.946	1.074	1.063	1.248
6		96	1.426	0.913	0.886	1.028	210.207	4.952	1.736	1.984	0.920	1.001
		144	1.014	1.017	1.086	1.050	67.452	1.770	1.079	1.116	1.120	0.984
		240	1.211	1.018	1.026	1.000	30.403	1.510	1.116	1.251	1.419	1.321
		480	0.905	1.148	1.210	1.107	9.284	1.447	1.048	1.160	1.312	1.262
8		96	1.267	0.768	0.807	0.875	184.654	3.934	1.550	1.665	0.980	0.857
		144	0.908	0.918	0.920	0.869	60.869	1.745	1.048	1.120	1.016	1.011
		240	0.986	1.030	0.943	0.968	20.176	1.047	0.931	1.003	0.942	0.915
		480	0.999	1.218	1.307	1.122	8.244	1.011	0.925	0.934	1.126	0.980
10		96	1.283	0.855	0.829	0.800	158.277	3.384	1.416	1.447	0.787	0.782
		144	0.805	0.826	0.730	0.719	56.227	1.712	0.951	0.994	0.809	0.847
		240	1.062	1.236	1.289	1.249	29.587	1.909	1.275	1.408	1.481	1.312
		480	0.939	1.008	1.157	1.036	8.927	1.378	1.187	1.267	1.080	1.210

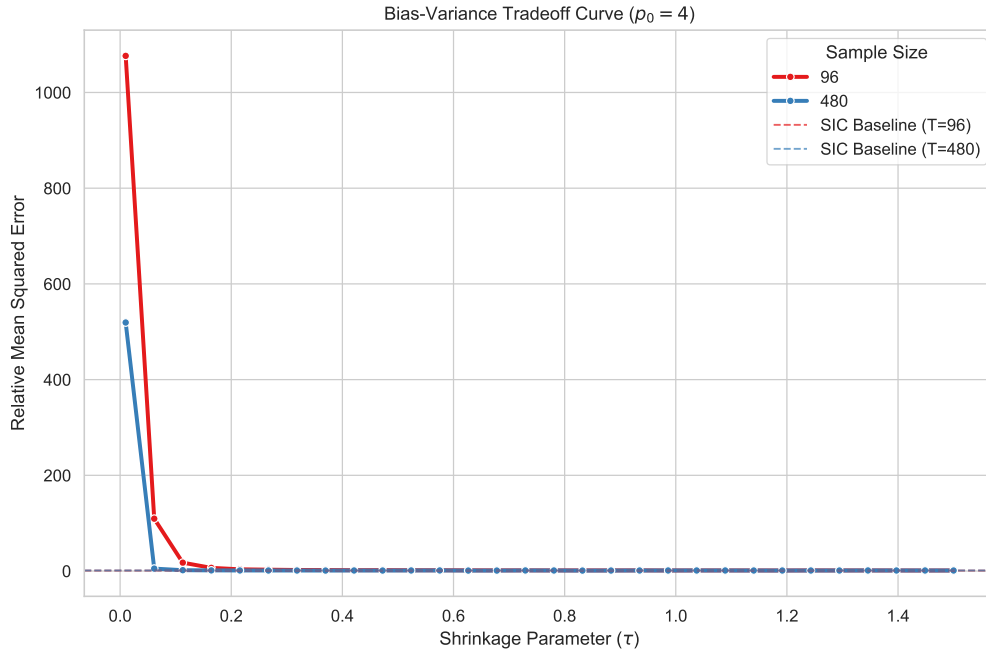


Figure 2: Bias-Variance Tradeoff Curve for SVAR Estimation. Demonstrates the out-of-sample forecasting error across varying degrees of shrinkage (τ) for small and large sample sizes.

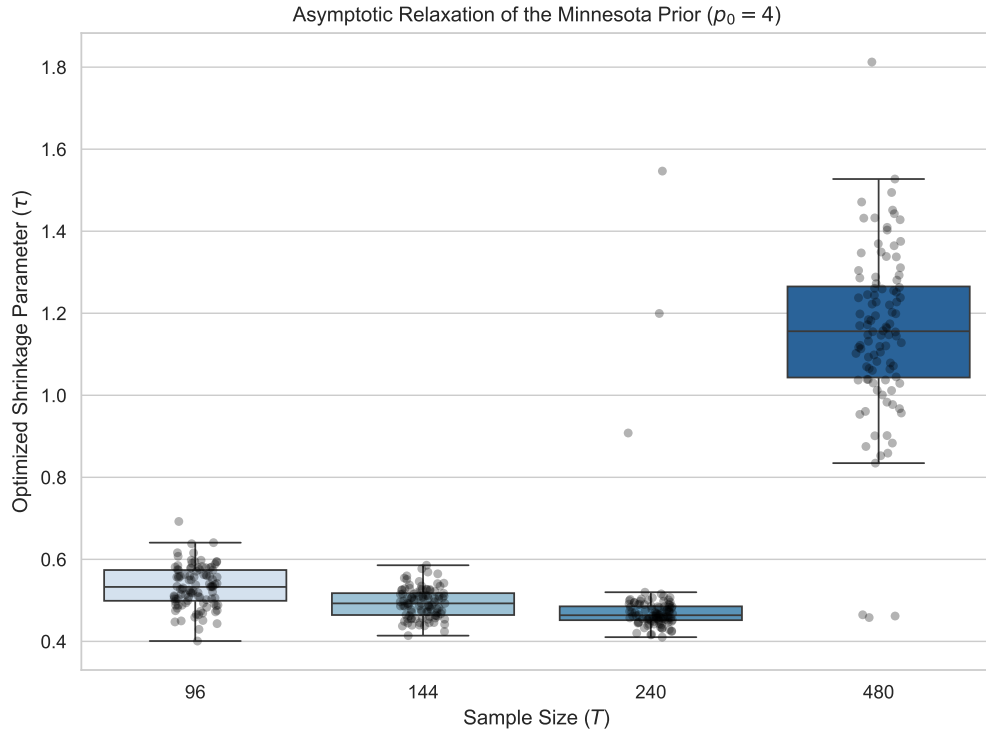


Figure 3: Asymptotic Relaxation of the Minnesota Prior. As the sample size (T) increases, the optimizer naturally selects looser priors, letting the data dominate the estimation.

3.3 The Uncertainty Strategy: Averaging vs. Selection

Here, we evaluate True Bayesian Model Averaging (BMA) against Information Criteria. We focus on how BMA acts as an insurance policy against catastrophic lag misspecification in small samples.

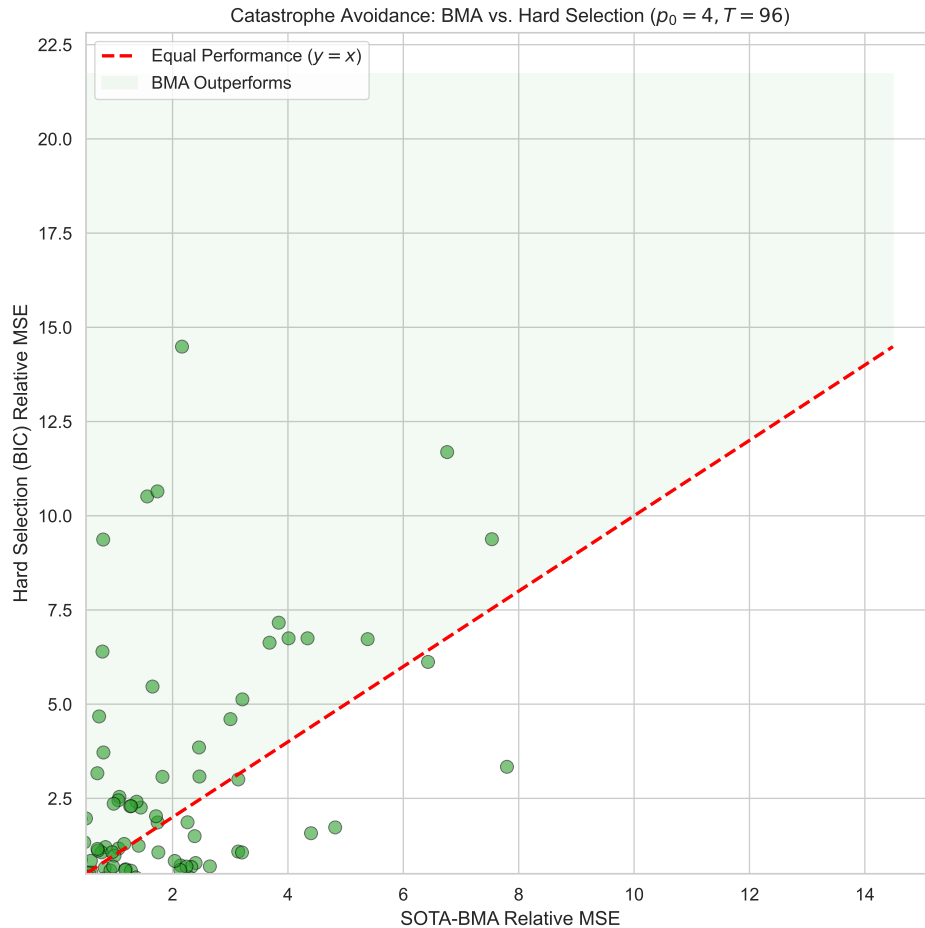


Figure 4: Catastrophe Avoidance: BMA vs. Hard Selection (BIC). Points above the 45-degree line indicate iterations where BMA outperformed BIC, notably smoothing out catastrophic tail risks.

Table 3: Relative Mean Squared Error: Hard Selection vs. Bayesian Model Averaging.

p_0	T	Estimator _{Mapped}	AIC	AICc	SIC (BIC)	HQC	OLS BMA (BIC-W)	Hybrid-BVAR (Fixed τ)	Hybrid-BMA (Fixed τ)	SOTA-BMA (Opt τ)
4	96		1.328	0.957	0.985	0.998	1.080	1.475	1.206	0.933
	144		1.106	1.206	1.145	1.085	1.165	1.962	2.043	1.245
	240		1.052	1.042	1.067	1.056	1.050	1.134	1.494	1.020
	480		0.956	1.589	1.214	1.091	0.978	1.428	0.782	1.248
6	96		1.426	0.913	0.886	1.028	0.744	1.574	1.456	1.001
	144		1.014	1.017	1.086	1.050	1.004	1.261	1.558	0.984
	240		1.211	1.018	1.026	1.000	1.253	1.475	1.265	1.321
	480		0.905	1.148	1.210	1.107	1.061	1.144	1.059	1.262
8	96		1.267	0.768	0.807	0.875	0.673	1.678	1.256	0.857
	144		0.908	0.918	0.920	0.869	0.985	1.168	1.018	1.011
	240		0.986	1.030	0.943	0.968	0.936	0.953	1.115	0.915
	480		0.999	1.218	1.307	1.122	0.985	1.328	1.069	0.980
10	96		1.283	0.855	0.829	0.800	0.839	1.272	1.276	0.782
	144		0.805	0.826	0.730	0.719	0.847	1.004	1.151	0.847
	240		1.062	1.236	1.289	1.249	1.761	1.139	2.212	1.312
	480		0.939	1.008	1.157	1.036	0.930	1.205	1.567	1.210

Table 4: Posterior Lag Weight Distribution for BMA ($p_0 = 4$).

Estimator T Statistic / Lag	Hybrid-BMA (OLS W) 96	SOTA-BMA (MDD W) 96	Hybrid-BMA (OLS W) 144	SOTA-BMA (MDD W) 480
$E[p]$	4.18	1.00	4.43	1.00
$p=1$	19.5	18.5	14.4	18.5
$p=10$	3.0	2.6	2.4	2.6

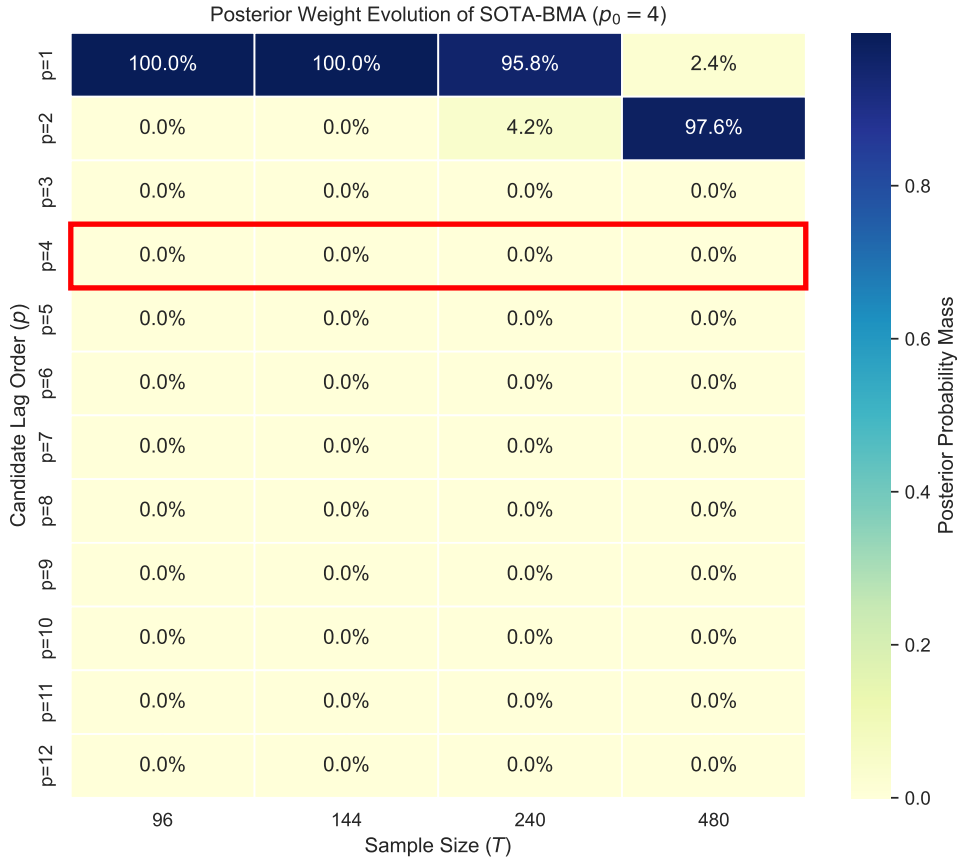


Figure 5: Posterior Weight Evolution of SOTA-BMA. At small sample sizes, probability mass is dispersed to hedge uncertainty. As T grows, the distribution concentrates precisely on the true DGP.

4 Conclusion

Summarize the main findings, specifically highlighting how optimal τ selection and true BMA systematically outperform standard econometric heuristics in finite samples.